Date: 8/08 /2025

Lab Practical #10:

Study of IP Addressing and sub-netting.

Practical Assignment #10:

1. Find default subnet masks, network bits, host bits, hosts per subnet, no of subnets, subnet number, 1st valid IP address, last valid IP address, and broadcast address.
   i. 8.1.4.5/16
   ii. 130.4.102.1/24
   iii. 130.4.102.1/22
   iv. 199.1.1.100/27
2. A host in a class C network has been assigned an IP address 192.168.17.9. Find the number of addresses in the block, the first address, and the last address.
3. An address in a block is given as 185.28.17.9. Find the number of addresses in the block, the first address, and the last address.
4. A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address, last address, number of addresses in a block?
5. Subnet the IP address 216.21.5.0 into 30 hosts in each subnet. Find Class, Default Mask, subnet mask, Bit Borrowed, New subnet mask, No. of Hosts & Subnet, Network Ranges (Subnets).
6. Subnet the IP address 192.10.20.0 into 52 hosts in each subnet. Find Class, Default Mask, Bit Borrowed, New subnet mask, No. of Hosts & Subnet, Network Ranges (Subnets).
7. Determining the Subnet mask for Devices A and B:
   a) Device A: 172.16.17.30/20
   b) Device B: 172.16.28.15/20

Date: 8/08 /2025

Page No. : 1

Date :

Assignement - 10

1) Find default subnet masks, host bits, hosts per subnet, no of ~~sb~~subnets, subnet number, 1st valid IP Address, last IP Address, broadcast add

i) 8.1.4.5/16

- class = A
- Default subnet mask - 255.0.0.0
- Network bits = 16
- Host bits = 16
- Hosts per subnet = 65,534
- No of subnets = 2^8 = 256
- subnet no. = 8.1.0.0
- 1st Valid = 8.1.0.1
- last Valid = 8.1.255.254
- Broadcast = 8.1.255.255

ii) 130.4.102.1/24

- class = B
- Default subnet mask - 255.255.0.0
- Network bits = 24
- Host bits = 8
- Host per subnet = 254
- No. of subnets = 2^8 = 256
- Subnet no. = 130.4.102.0
- 1st valid = 130.4.102.1
- last valid = 130.4.102.254
- Broadcast = 130.4.102.255

Darshan UNIVERSITY

Date: 8/08 /2025

Page No. : 2

Date :

iii) 130.4.102.1/22

- Class = B
- Default subnet mask = 255.255.0.0
- Network bits = 22
- Host bits = 10
- Hosts per subnet = 1022
- No. of ~~student~~ subnets = 2^6 = 64
- subnet number = 130.4.100.0
- 1st Valid = 130.4.100.1
- Last Valid = 130.4.103.254
- Broad Cast = 130.4.103.255

iv) 199.1.1.100/27

- Class = C
- Default subnet mask = 255.255.255.0
- Network bits = 27
- Host bits = 5
- Host per subnet = 30
- No. of subnets = 2^3 = 8
- subnet no. = 199.1.1.96
- 1st valid = 199.1.1.97
- last Valid = 199.1.1.126
- Broadcast = 199.1.1.127

Darshan UNIVERSITY

Date: 8/08 /2025

Page No. : 3
Date :

2) A host in class C network has been assigned an IP address 192.168.17.9. Find the no. of address in the block, the first address and the last address.

- Block : 24 - 255.255.255.0
- no. of address = $2^{32-24} = 2^8 = 256$
- First Address = 192.168.17.0
- Last Address = 192.168.17.255

3) An address in a block is given as 185.28.17.9 Find the number of address in the block, the first address, and last address.

- Block - 16 255.255.0.0
- Number of address - 65,536
- First address - 185.28.0.0
- last address - 185.28.255.255

4) A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first addresses, last address, number of add. in block

- Subnet mask = 255.255.255.240
- Block Size = $2^{32-28} = 16$
- Fourth - octet of block = 16 => cover 32-47
- First address = 205.16.37.32
- last address = 205.16.37.47
- No. of address in the block = 16

Darshan UNIVERSITY

Date: 8/08 /2025

Page No.: 4

Date:

5) Subnet the IP Address 216.21.5.0 into 30 hosts in each subnet. Find class, Default Mask, subnet mask, Bit Borrowed, New subnet mask, No. of Hosts & subnet, Network Ranges.

- Class = C
- Default subnet mask = 255.255.255.0
- Bit Borrowed = 3
- New Subnet mask = 255.255.255.224
- No. of subnets = 2^3 = 8
- Host per subnet = $2^5 - 2 = 30$

-> Network Range:

1) 216.21.5.0 - 216.21.5.31
2) 216.21.5.32 - 216.21.5.63
3) 216.21.5.64 - 216.21.5.95
4) 216.21.5.96 - 216.21.5.127
5) 216.21.5.128 - 216.21.5.159
6) 216.21.5.160 - 216.21.5.191
7) 216.21.5.192 - 216.21.5.223
8) 216.21.5.224 - 216.21.5.255

Date: 8/08 /2025

Page No. 5
Date :

6) Subnet the IP Address 192.10.20.0 into 52 hosts in each subnet. Find class, Default Mask, Bit Borrowed, New subnet mask, No. of Hosts & Subnet, Network Ranges.

- class = c
- Host Bit Borrowed = 2
- New subnet mask = 255.255.255.192
- No. of Subnet = $2^2 = 4$
- Host per subnet = 62

Network Ranges :
1) 192.10.20.0 - 192.10.20.13
2) 192.10.20.14 - 192.10.20.127
3) 192.10.20.128 - 192.10.20.191
4) 192.10.20.192 - 192.10.20.255

7) Determining the subnet mask for Device A & B.

a) Device A = 172.16.17.30/20
- Thired Octel = 16
- Network = 172.16.16.0
- Usable = 172.16.16.1 - 172.16.31.254
- Broadcast = 172.16.31.255

b) Device B = 172.16.28.15/20
- Network = 172.16.16.0
- Usable = 172.16.16.1 - 172.16.31.254
- Broadcast - 172.16.31.255
- Thired octel - 16

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